

A Technical Introduction to Image to image translation

Section 1

$$\frac{\partial^2 f(x, y)}{\partial x^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\frac{\partial^2 f(x, y)}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

Section 2

Optimization algorithms Each optimization algorithm is an adaptation of models from a variety of fields and they are set apart by their unique cost functions. The common trait of cost functions is to penalize change in pixel value as well as difference in pixel label when compared to labels of neighboring pixels.

Section 3

$$P(\lambda \mid f) = P(f \mid \lambda) P(\lambda) \sum_{\lambda \in \Lambda} P(f \mid \lambda) P(\lambda)$$

$$\frac{P(f_i \mid \lambda)}{P(\lambda)} = \frac{P(f_i \mid \lambda)}{P(\lambda)} \sum_{\lambda \in \Lambda} P(f_i \mid \lambda) P(\lambda)$$

Section 4

The functional value is the sum of the total length of the segmentation curve K , the smoothness of the approximation u , and its distance to the original image f . The weight of the smoothness penalty is adjusted by $\mu > 0$. The Potts model is often called piecewise constant Mumford-Shah model as it can be seen as the degenerate case $\mu \rightarrow \infty$. The optimization problems are known to be NP-hard in general but near-minimizing strategies work well in practice. Classical algorithms are graduated non-convexity and Ambrosio-Tortorelli approximation.

Section 5

Alternative algorithms A range of other methods exist for solving simple as well as higher order MRFs. They include Maximization of Posterior Marginal, Multi-scale MAP estimation, Multiple Resolution segmentation and more. Apart from likelihood estimates, graph-cut using maximum flow and other highly constrained graph based methods exist for solving MRFs.

Section 6

Region-growing methods Region-growing methods rely mainly on the assumption that the neighboring pixels within one region have similar values. The common procedure is to compare one pixel with its neighbors. If a similarity criterion is satisfied, the pixel can be set to belong to the same cluster as one or more of its neighbors. The selection of the similarity criterion is significant and the results are influenced by noise in all instances. The method of

Statistical Region Merging (SRM) starts by building the graph of pixels using 4-connectedness with edges weighted by the absolute value of the intensity difference. Initially each pixel forms a single pixel region. SRM then sorts those edges in a priority queue and decides whether or not to merge the current regions belonging to the edge pixels using a statistical predicate. One region-growing method is the seeded region growing method. This method takes a set of seeds as input along with the image. The seeds mark each of the objects to be segmented. The regions are iteratively grown by comparison of all unallocated neighboring pixels to the regions. The difference between a pixel's intensity value and the region's mean, δ , is used as a measure of similarity. The pixel with the smallest difference measured in this way is assigned to the respective region. This process continues until all pixels are assigned to a region. Because seeded region growing requires seeds as additional input, the segmentation results are dependent on the choice of seeds, and noise in the image can cause the seeds to be poorly placed. Another region-growing method is the unseeded region growing method. It is a modified algorithm that does not require explicit seeds. It starts with a single region A_1 —the pixel chosen here does not markedly influence the final segmentation. At each iteration it considers the neighboring pixels in the same way as seeded region growing. It differs from seeded region growing in that if the minimum δ is less than a predefined threshold T then it is added to the respective region A_j . If not, then the pixel is considered different from all current regions A_i and a new region A_{n+1} is created with this pixel. One variant of this technique, proposed by Haralick and Shapiro (1985), is based on pixel intensities. The mean and scatter of the region and the intensity of the candidate pixel are used to compute a test statistic. If the test statistic is sufficiently small, the pixel is added to the region, and the region's mean and scatter are recomputed. Otherwise, the pixel is rejected, and is used to form a new region. A special region-growing method is called λ -connected segmentation (see also λ -connectedness). It is based on pixel intensities and neighborhood-linking paths. A degree of connectivity (connectedness) is calculated based on a path that is formed by pixels. For a certain value of λ , two pixels are called λ -connected if there is a path linking those two pixels and the connectedness of this path is at least λ . λ -connectedness is an equivalence relation. Split-and-merge segmentation is based on a quadtree partition of an image. It is sometimes called quadtree segmentation. This method starts at the root of the tree that represents the whole image. If it is found non-uniform (not homogeneous), then it is split into four child squares (the splitting process), and so on. If, in

contrast, four child squares are homogeneous, they are merged as several connected components (the merging process). The node in the tree is a segmented node. This process continues recursively until no further splits or merges are possible. When a special data structure is involved in the implementation of the algorithm of the method, its time complexity can reach $O(n \log n)$, an optimal algorithm of the method.

Section 7

Related images such as a photo album or a sequence of video frames often contain semantically similar objects and scenes, therefore it is often beneficial to exploit such correlations. The task of simultaneously segmenting scenes from related images or video frames is termed co-segmentation, which is typically used in human action localization. Unlike conventional bounding box-based object detection, human action localization methods provide finer-grained results, typically per-image segmentation masks delineating the human object of interest and its action category (e.g., Segment-Tube). Techniques such as dynamic Markov Networks, CNN and LSTM are often employed to exploit the inter-frame correlations.